ATM Simulator - Exploratory Testing Charter

Charter for the 30-minute

1. Charter Header

Charter ID	ATM-ET-001	**Date**	2026-10-20
Tester	Eng \ Belal Ahmed Ashor	**Feature / Area**	Core ATM transaction flow
Build / Version	Web version / build number not provided	**Environment**	Desktop PC - Google Chrome 155
Session Duration	30 minutes	**Session Type**	Functional exploratory testing

2. Mission and Scope

2.1 Mission Statement

Explore: the ATM Simulator core flow, from account identification and PIN authentication through balance inquiry, cash deposit, cash withdrawal, Mini Statement, Change PIN, and exit / re-login.

To discover: unexpected behavior in input validation, transaction calculations, navigation, account state, PIN handling, and transaction history during normal, invalid, and boundary-value scenarios.

And report: reproducible defects, confusing or inconsistent feedback, and coverage gaps that could affect transaction correctness or user confidence.

2.2 Scope

In Scope	Out of Scope
- Account Number input and validation, including empty, invalid numeric, alphabetic, special-character, emoji, and length cases. - Valid and invalid PIN entry, remaining-attempt messages, and switching between accounts. - Check Balance, Cash Deposit, Cash Withdrawal, including boundary values, decimals, insufficient funds, and full-balance withdrawal. - Mini Statement content and timestamps, transaction navigation, Change PIN same-value, mismatch and exit and re-login.	- Load, stress, and performance testing. - Penetration testing or exhaustive security assessment. - Backend, database, or API verification. - Cross-browser, mobile-device, accessibility, and extended-session testing.

2.3 Starting Point and Preconditions

Application URL	https://atmsimulatorsystem.vercel.app/
Starting state	Open the application landing page and begin with the demo account number displayed in the application. Use the demo PIN 1234 for the valid authentication path.
Test data	Account Number: empty, invalid numeric values, letters / special characters / emoji, and the displayed demo account. PIN: valid and invalid entries. Deposit: 1 and 1.5 among other numeric values. Withdrawal: 0, 99, 99.9, 100, 200, 250, decimal values, and amounts above / equal to the available balance. Change PIN: same value, mismatch, letters, and special characters.

3. Testing Heuristics and Approaches

3.1 Risks and Concerns

Risk area	Question to investigate
Transaction navigation	Can a user move between transaction screens in an unexpected order, and is the resulting state consistent with the intended workflow?
Transaction timestamps	Do displayed dates and times remain reliable if the PC system time changes, and does the history communicate transaction order accurately?
Change PIN and login validation	Do both screens accept and reject the same PIN character types, and does a successful PIN change leave the account in a usable state?
PIN attempts across accounts	Does the remaining-attempt message reflect the currently selected account and update at the correct time?
Transaction limits and history	Are invalid amounts rejected without changing the balance, and are successful transactions reflected correctly in the Mini Statement?

3.2 Testing Heuristics (SFDPOT)

Heuristic	How it applies to this session
Structure	Inspect the ATM screen, input controls, transaction screens, and navigation options.
Function	Check that authentication, balance inquiry, deposit, withdrawal, Mini Statement, Change PIN, and exit perform their visible functions.
Data	Vary empty, invalid, boundary, decimal, and repeated values; compare valid and invalid PIN formats.
Platform	Observe behavior in the recorded desktop environment and Google Chrome version.
Operations	Vary the order of transactions, move between screens, switch accounts, exit, and log in again.
Time	Check transaction timestamps and remaining-attempt messages after repeated actions or account changes.

3.3 Testing Approaches

Approach	Planned exploration
Happy path	Complete valid login, balance inquiry, deposit, withdrawal, statement review, Change PIN validation, and exit / re-login.
Negative testing	Try missing / invalid account values, incorrect PINs, mismatched PIN confirmation, and values that should be rejected.
Boundary values	Explore account-number length, low deposit values, withdrawal values around the displayed minimum of 100, and insufficient / full-balance cases.
State and sequence	Navigate between transactions, inspect the statement before and after transactions, change the system time, and switch accounts.
Consistency checks	Compare field validation between Change PIN and login, and reconcile successful transaction amounts with balance and statement changes.

3.4 Oracles - How to Judge the Result

- The balance should change by the amount of a successful deposit or withdrawal and should not change after a rejected transaction.
- Successful transactions should appear in the Mini Statement with information consistent with the action performed.
- Invalid inputs should be handled predictably, with clear feedback and without unintended state changes.
- PIN creation / change and subsequent login should use compatible validation rules; attempt messages should describe the current account state.